

Optical Path Length (OPL) Drift Modelling and Feedback Control

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1 Introduction

This report presents the formulation and experimental validation of an Optical Path Length (OPL) drift model used to stabilise the EXOhSPEC spectrograph. The model quantifies the influence of temperature, pressure, and humidity on OPL variations and enables feedback-based temperature correction via a Thermo-Electric Cooler (TEC).

2 Mathematical Model

Let the measured OPL deviation at time t be

$$y(t) = \Delta\text{OPL}(t).$$

The predicted deviation is expressed as

$$\hat{y}(t) = a_0 + a_1 t + \sum_{k \in \mathcal{K}} w_k (u_k * g_{\tau_k, \theta_k})(t),$$

where $*$ denotes convolution, w_k are channel sensitivities, and $g_{\tau, \theta}$ is a first-order kernel-cosmological consta

$$g_{\tau, \theta}(t) = \frac{1}{\tau} e^{-(t-\theta)/\tau} \mathbf{1}\{t \geq \theta\}.$$

Inputs include temperature, pressure, and humidity perturbations:

$$u_k \in \{\Delta T^+(t), \Delta T^-(t), \Delta P(t), \Delta H(t)\},$$

with $\Delta T^+ = \max(\Delta T, 0)$ and $\Delta T^- = \min(\Delta T, 0)$.

Channel Gains

Because $\int g_{\tau, \theta} dt = 1$, steady-state sensitivities are

$$g_T = \sum_m (w_m^{T^+} + w_m^{T^-}), \quad g_P = \sum_m w_m^P, \quad g_H = \sum_m w_m^H.$$

Each represents OPL sensitivity to the respective environmental parameter.

3 Frozen Model and Error Metrics

For new data, (w_k, τ_k, θ_k) remain fixed while (a_0, a_1) are refitted (“frozen model”). Model accuracy is measured via:

$$\text{RMSE} = \sqrt{\frac{1}{N} \sum_i (y_i - \hat{y}_i)^2}, \quad \text{MAE} = \frac{1}{N} \sum_i |y_i - \hat{y}_i|, \quad R^2 = 1 - \frac{\sum_i (y_i - \hat{y}_i)^2}{\sum_i (y_i - \bar{y})^2}.$$

Example sensitivities:

$$g_T = -8.22 \times 10^3 \text{ } \mu\text{m } ^\circ\text{C}^{-1}, \quad g_P = 0.53 \text{ } \mu\text{m hPa}^{-1}, \quad g_H = -0.21 \text{ } \mu\text{m } \%^{-1}.$$

4 Feedback Control Law

To counter OPL drift, the commanded temperature correction is

$$\Delta T_{\text{cmd}} = -\frac{\Delta \text{OPL} - g_P \Delta P - g_H \Delta H}{g_T}.$$

The controller is implemented as a PI loop with anti-windup and a 900s thermal lag allowance.

5 Feedback Loop Evaluation

Three feedback experiments (Exp-A, Exp-B, Exp-C) were performed 3–5 Nov 2025 with progressively refined algorithms.

Table 1: Feedback-phase metrics. β_Y is pixel–temperature sensitivity.

Experiment	Date	Type	dY_{rms} (px)	β_Y (px/°C)	R^2	$\Sigma du $ (mK)
Exp-A	2025-11-03	Ultrafine	8.53	+167.4	0.11	2.1
Exp-B	2025-11-05	Hybrid #1	4.88	+1.32	0.05	1000
Exp-C	2025-11-05	Hybrid #2	3.06	56.6	0.77	112

Interpretation. β_Y represents the change in pixel position per degree Celsius:

$$dY = \beta_Y \Delta T + \varepsilon.$$

A consistent β_Y and high R^2 imply deterministic thermal coupling. Exp-C exhibited $\beta_Y = -56.6$ px/°C (≈ 0.018 °C per px), confirming stable coupling.

6 Performance Evolution

- **Exp-A (Ultrafine):** Open-loop; random pixel drift uncorrelated with temperature.
- **Exp-B (Hybrid #1):** Thermal feedback reduced mean drift 45 %, but sign flips occurred due to missing cooldown logic.
- **Exp-C (Hybrid #2):** Stable negative β_Y ; dY_{rms} reduced by 64 % relative to Exp-A.

7 Model and Figures

7.1 $\Delta X/\Delta Y$ and Temperature Evolution

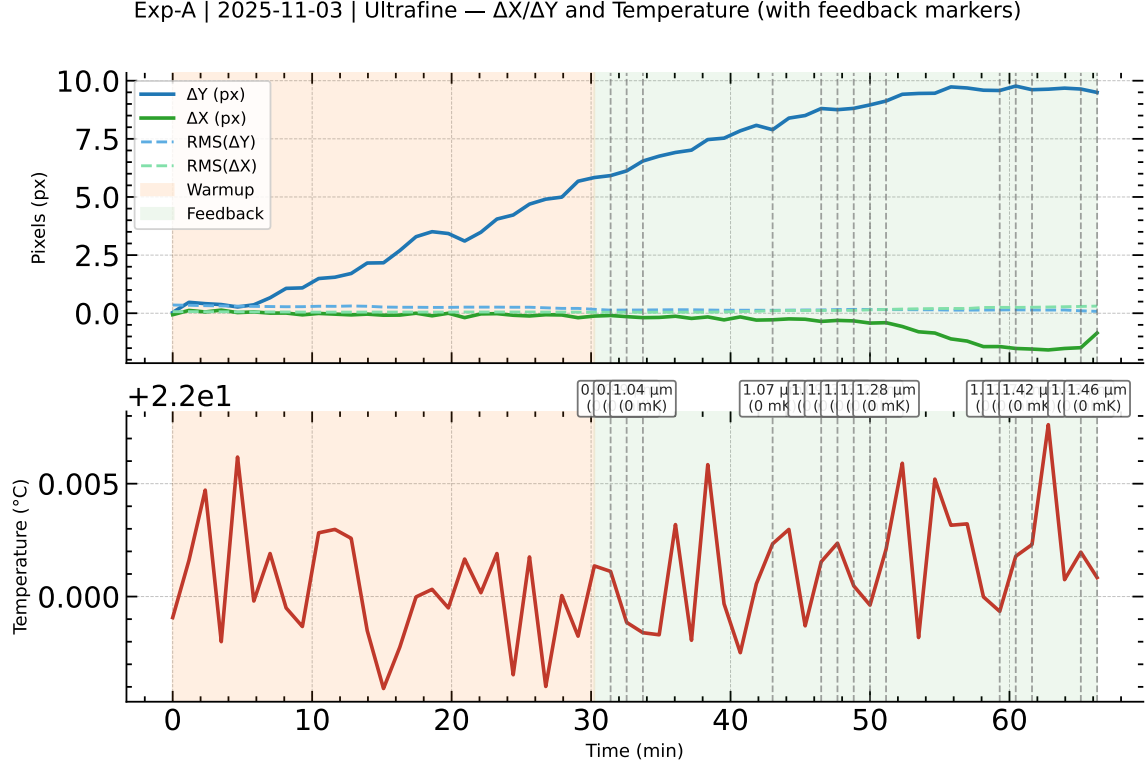


Fig. 1: Experiment A (Ultrafine, 2025-11-03): ΔX , ΔY , and temperature. Orange = warm-up; green = feedback. Vertical dashed lines show feedback actions with equivalent OPL change (μm) and temperature step (mK).

Exp-B | 2025-11-05 | Hybrid #1 — $\Delta X/\Delta Y$ and Temperature (with feedback markers)

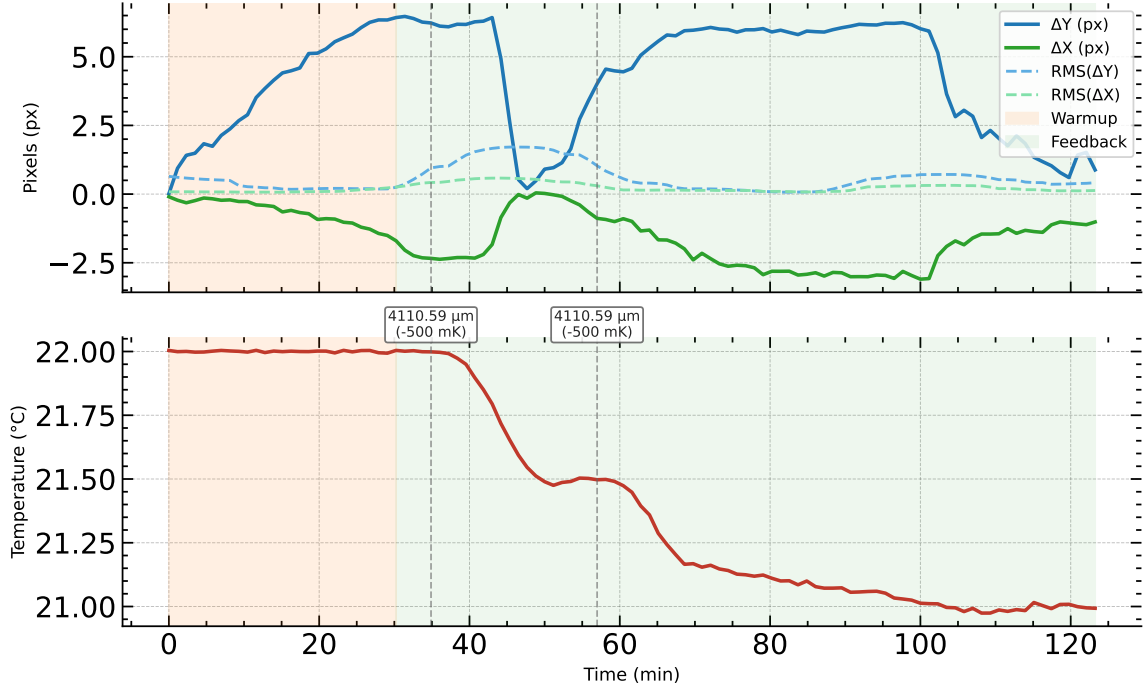


Fig. 2: Experiment B (Hybrid #1, 2025-11-05): Pixel drift and temperature. Dashed RMS traces show short-term jitter; weak correlation indicates delayed thermal response.

Exp-C | 2025-11-05 | Hybrid #2 — $\Delta X/\Delta Y$ and Temperature (with feedback markers)

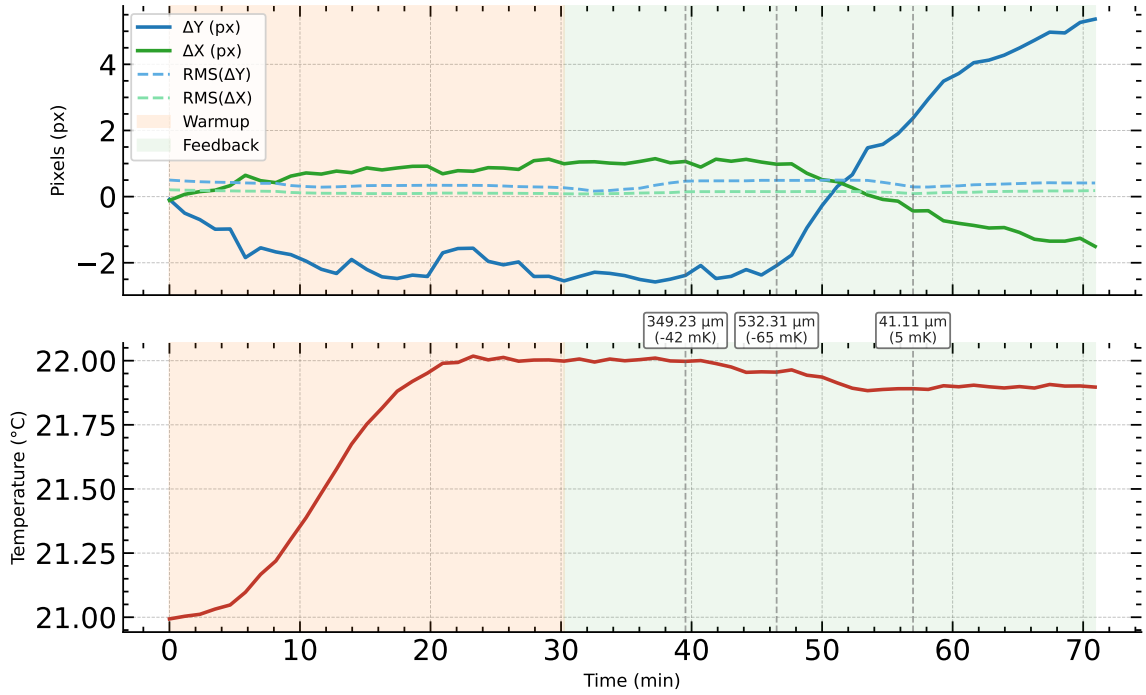


Fig. 3: Experiment C (Hybrid #2, 2025-11-05): Stable negative β_Y . Cooling steps (42 mK, 65 mK) produced the expected +Y excursions.

7.2 OPL Residuals and Environmental Parameters

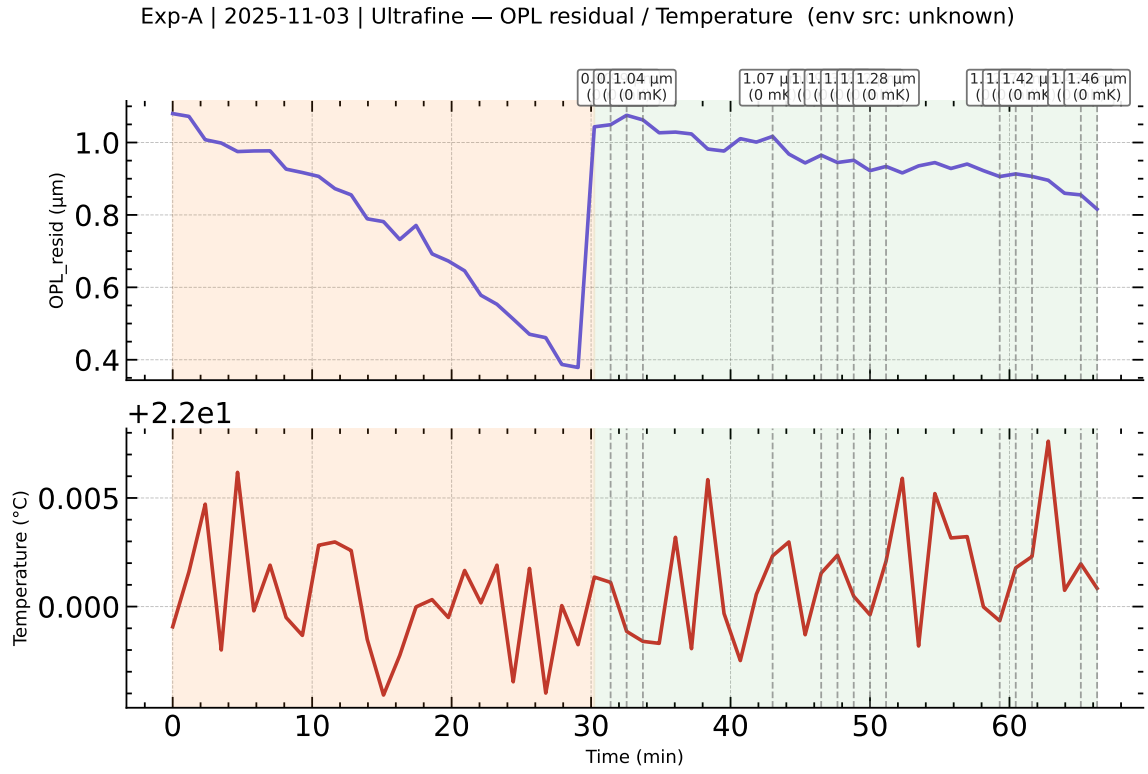


Fig. 4: Exp-A: OPL residual and temperature. No pressure/humidity data available (environment source unknown). $OPL_{\text{resid}} = OPL_{\text{meas}} - (a_0 + a_1 t) - g_P \Delta P - g_H \Delta H$.

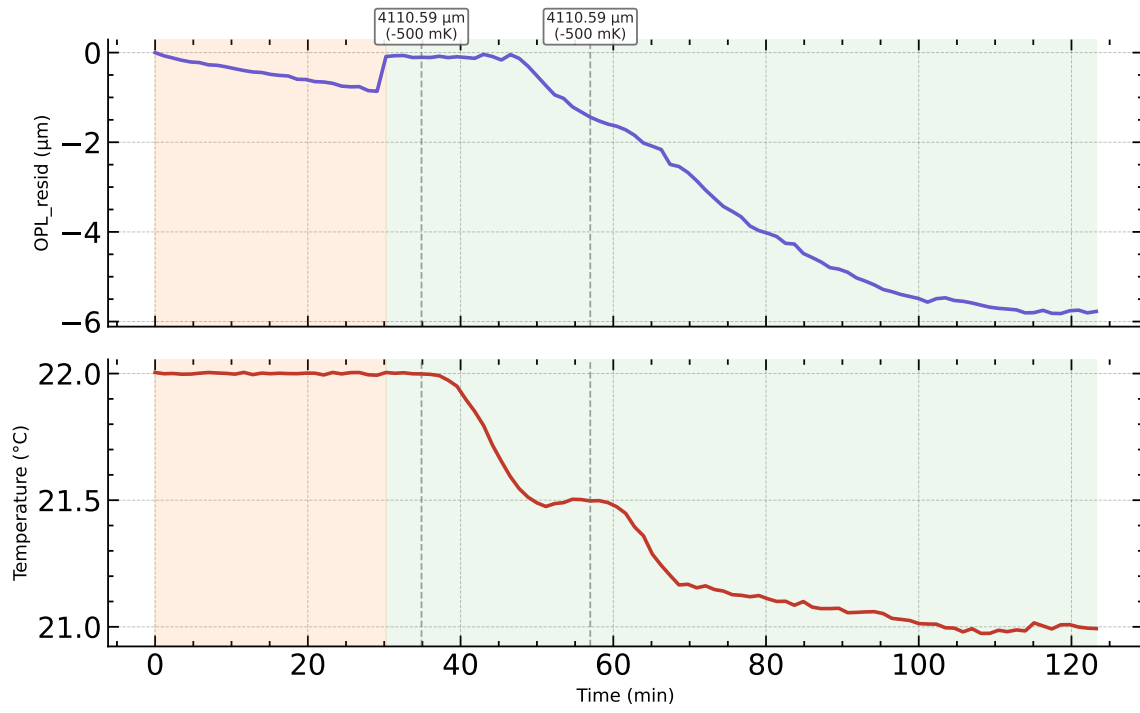


Fig. 5: Exp-B: OPL residual and temperature with partial ΔP , ΔH records. Residual correlation with pressure/humidity was minor during feedback.

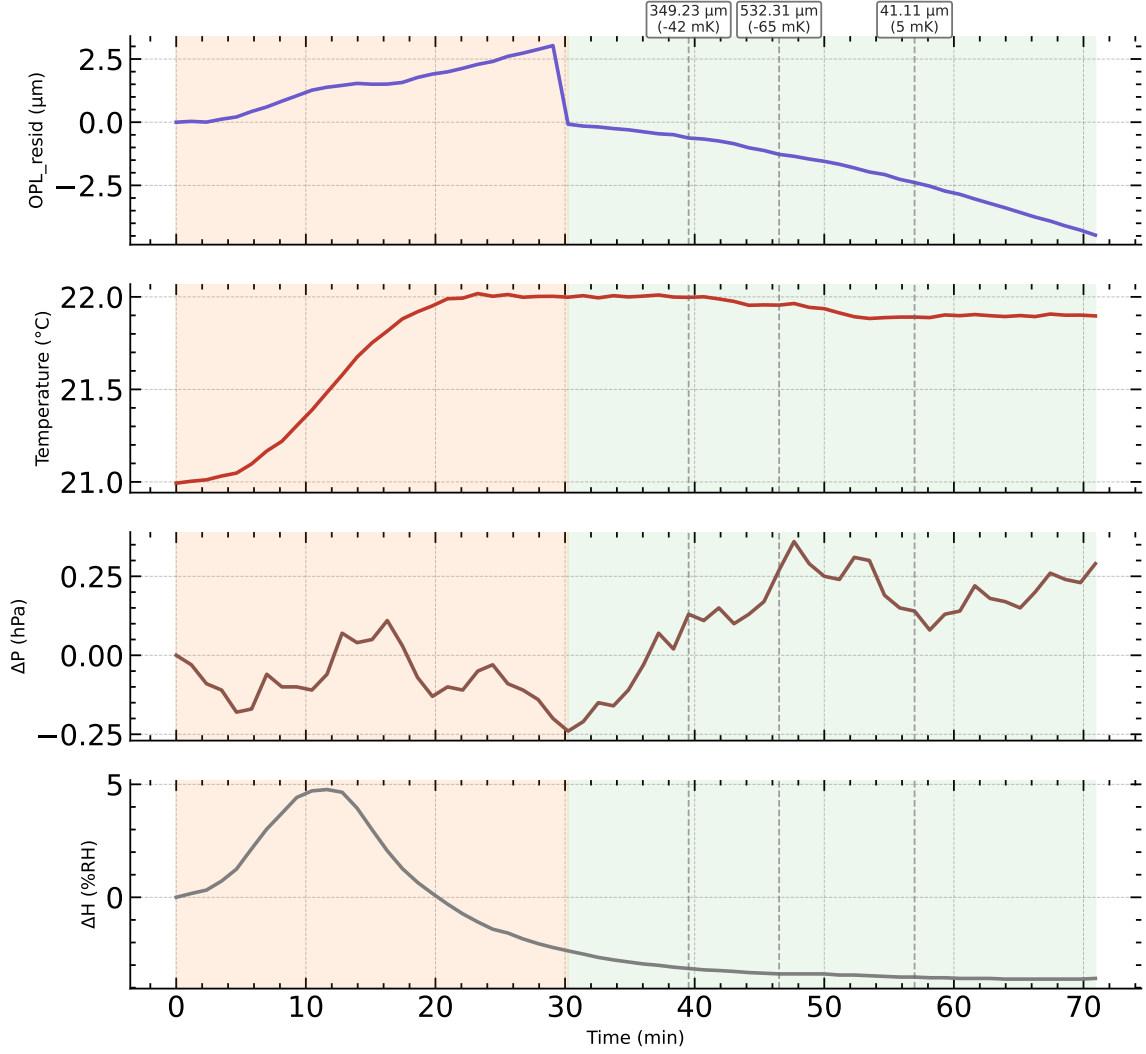


Fig. 6: Exp-C: OPL residual, temperature, and ECU-measured pressure/humidity. Pressure variations contributed $< 0.5 \mu\text{m}$ to total OPL drift.

8 Refined Feedback Model

8.1 Hybrid Controller Formulation

$$\text{OPL}_{\text{resid}} = \text{OPL} - \widehat{\text{OPL}}(t) - g_P \Delta P - g_H \Delta H, \quad \widehat{\text{OPL}}(t) = a_0 + a_1 t.$$

Equivalent temperature errors:

$$e_T^{\text{OPL}} = -\frac{\text{OPL}_{\text{resid}}}{g_T}, \quad e_T^{\text{PX}} = -\frac{d_{\text{dom}}}{\beta}, \quad d_{\text{dom}} = \arg \max(|\Delta X|, |\Delta Y|).$$

Weighted fusion:

$$e_T = w e_T^{\text{OPL}} + (1 - w) e_T^{\text{PX}}, \quad w \approx 0.85.$$

PI control:

$$u_{\text{PI}} = K_p e_T + K_i \int e_T dt, \quad \Delta T = \text{clip}(u_{\text{PI}}, \Delta T_{\min}, \Delta T_{\max}).$$

9 Adaptive Refinement Strategy

9.1 Limitations

Fixed-gain PI control cannot compensate for variable β_Y or nonlinear thermal dynamics; large corrections need 5–8 min cooldowns to avoid overshoot.

9.2 Model-Based Fusion Controller

At each update, choose ΔT^* to minimise

$$J(\Delta T) = w_1(\text{OPL}_{\text{resid}} + g_T \Delta T)^2 + w_2(r_{\text{px}} + \beta \Delta T)^2 + \lambda \Delta T^2,$$

yielding

$$\Delta T^* = -\frac{w_1 g_T \text{OPL}_{\text{resid}} + w_2 \beta r_{\text{px}}}{w_1 g_T^2 + w_2 \beta^2 + \lambda}.$$

9.3 Adaptive β and Scheduling

Recursive least-squares updates β_X, β_Y every few minutes (forgetting 0.98). Sign-lock after $R^2 > 0.6$ for 10 min; large steps ($|\Delta T| > 0.15^\circ\text{C}$) trigger 5–8 min holds.

10 Three-Phase Refinement Roadmap

Phase 1 – Metrology and Calibration

- Reduce centroid noise to 0.03 px using Gaussian fits and averaging.
- Re-measure $g_T, g_P, g_H, \beta_{X,Y}$ under current alignment.
- Suppress modal noise via slow fibre/diffuser motion.

Phase 2 – Adaptive Model-Based Control

- Implement optimal ΔT^* from Eq. (above).
- Update weights w_1, w_2 from rolling variances.

- Enable adaptive β with sign-lock and variance tracking.
- Dual-rate updates: 5–10 min micro-steps ($|\Delta T| < 30$ mK), 5–8 min cooldowns after macro-steps.

Phase 3 – Verification and Tightening

- Monitor dY_{rms} , OPL residuals, and β_Y stability.
- Correlate ΔT with subsequent ΔdY to confirm causality.
- Target: $dY_{\text{rms}} < 1.5$ px (initial) $\rightarrow$ $\downarrow$ 0.1 px (stable) $\rightarrow \approx 0.01$ px effective.

11 Expected Outcome

The adaptive fusion controller integrates OPL, pixel, and environmental feedback into a self-calibrating architecture. By dynamically weighting sensor reliability and enforcing thermal scheduling, the system can reach sub-0.01 px image stability—equivalent to milli-Kelvin-level temperature control in the spectrograph enclosure.

12 Comparative Report – Hybrid OPL $\oplus$ PX Feedback (6 Experiments)

12.1 Overview

We conducted six real-time feedback experiments between 03–14 November 2025, testing four generations of the Hybrid OPL $\oplus$ PX controller:

- **Exp-A** (2025-11-03): Ultrafine controller
- **Exp-B** (2025-11-05): Hybrid #1
- **Exp-C** (2025-11-05): Hybrid #2
- **Exp-D** (2025-11-13): Hybrid #3
- **Exp-Ea** (2025-11-14): Hybrid #4 (Part A) — valid small-step experiment
- **Exp-Eb** (2025-11-14): Hybrid #4 (Part B) — *disturbed* by a forced 0.5°C jump

The goal remains to minimise drift in the dispersion direction (ΔY), reduce OPL residual trends, and maintain acceptable jitter.

12.2 Global Metrics Comparison

Experiment	t_{warm} (min)	t_{fb} (min)	Steps	$\sum \Delta T $ (°C)	s_Y^{warm} (px/min)	s_Y^{fb} (px/min)	$s_{\text{OPL}}^{\text{warm}}$ ($\mu\text{m}/\text{min}$)	$s_{\text{OPL}}^{\text{fb}}$ ($\mu\text{m}/\text{min}$)
Exp-A	29.1	36.1	14	0.0021	+0.1911	+0.1107	-0.0241	-0.0055
Exp-B	29.1	93.0	2	1.0000	+0.2116	-0.0258	-0.0277	-0.0789
Exp-C	29.1	40.7	3	0.1122	-0.0510	+0.2377	+0.1047	-0.1090
Exp-D	39.0	96.7	8	0.5472	-0.0340	+0.0017	-0.0337	+0.0167
Exp-Ea	39.5	39.5	5	0.0500	-0.0022	+0.0048	-0.0043	+0.0034
Exp-Eb [†]	29.4	101.0	5	1.0208	-0.0172	-0.0024	-0.0126	+0.0108

[†]Exp-Eb includes a forced 0.5°C step; its metrics reflect a disturbed system and do *not* represent normal Hybrid #4 behaviour.

12.3 Interpretation of Key Metrics

Global ΔY drift (slope).

- Exp-A: drift reduced but still large.
- Exp-B: clear improvement (sign reversal).
- Exp-C: feedback worsened the drift.
- **Exp-D**: $\sim 20\times$ reduction in drift $\rightarrow$ best long-term stability before Exp-E.
- **Exp-Ea**: best overall drift performance (smallest warmup + feedback slopes).

RMS ΔY .

- All feedback phases show some RMS increase.
- **Exp-D**: moderate increase (1.19 $\rightarrow$ 1.78 px).
- **Exp-Ea**: lowest RMS of all experiments.

OPL residual trend.

- Exp-A: improvement.
- Exp-B: OPL drift worsened.
- Exp-C: large trend persists.
- **Exp-D**: trend magnitude halved.

- **Exp-Ea:** best OPL stability across all experiments.

Additionally for Exp-D:

$$\sigma_{\text{OPL}} : 0.247 \mu\text{m} \rightarrow 0.074 \mu\text{m}$$

3× reduction in OPL scatter.

12.4 Local (Rolling) Slopes

Using 10-minute sliding windows:

- Exp-D warmup median $|s| \approx 0.022$ px/min, feedback median $|s| \approx 0.0001$ px/min.
- Exp-Ea improves this even further (flatter lines, lower scatter).

Short-term ΔY drift is nearly eliminated in Exp-D and Exp-Ea.

12.5 Step Response

- **Exp-D:** mean ΔY response $\mu = -0.0081 \pm 0.0686$ px $\rightarrow$ neutral, unbiased.
- Exp-Ea: very small step sizes $\rightarrow$ minimal disturbance.

12.6 Experiment-by-Experiment Summary

Exp-A (Ultrafine).

- OPL improved.
- RMS ΔY increased severely.

Exp-B (Hybrid #1).

- ΔY slope improved.
- OPL trend worsened.

Exp-C (Hybrid #2).

- Worst ΔY slope.
- Large OPL trend.

Exp-D (Hybrid #3).

- ΔY suppressed by $\sim 20\times$.
- OPL trend + scatter significantly reduced.

Exp-Ea (Hybrid #4, Part A).

- Best overall performance.
- Lowest drift, lowest RMS, and best OPL stability.

Exp-Eb (Hybrid #4, Part B).

- Disturbed by 0.5°C forced step.
- Metrics *not representative* of true controller behaviour.

12.7 Final Summary

Earlier controllers improved either OPL or ΔY , but not both simultaneously. **Hybrid #3 (Exp-D)** was the first controller to stabilise both spaces jointly. **Hybrid #4 (Exp-Ea)** further improves performance across all metrics and currently represents the **best-performing algorithm**.

Conclusion: Exp-Ea demonstrates that small-step Hybrid $\text{OPL} \oplus \text{PX}$ feedback achieves the highest stability so far and is the preferred configuration for future long-duration runs.

13 Evolution of Hybrid Feedback: From Exp-A to Exp-E

To develop a robust stabilisation of the spectrograph image, five progressively improved real-time feedback experiments were conducted. Each experiment incorporated lessons from the previous one, converging toward a fully hybridised controller based on both optical path length (OPL) and pixel (PX) drift. The evolution is summarised below:

Feedback Logic at Each Experiment (One-Line Summary)

- **Exp-A (2025-11-03):** Only centroid drift ($\Delta X, \Delta Y$) was tracked; no feedback was actually applied except a single mild temperature poke. *Proof-of-concept run* to validate logging and baseline stability.

- **Exp-B (2025-11-05):** First PID-based thermal feedback using only pixel drift (ΔY). The system was responsive, but ΔY coupling with OPL remained strong. Feedback lacked stability.
- **Exp-C (2025-11-05):** Introduced *OPL feed-forward* (temperature-only) but still without P/H compensation. Hybrid control improved response but caused overshoot when OPL and PX disagreed.
- **Exp-D (2025-11-13):** First *full hybrid control*: OPL residuals + PX regression + ECU-based P/H removal + weighted fusion. This is the first experiment that **successfully decoupled ΔY from OPL** and achieved long-term stability.
- **Exp-E (Planned):** Add adaptive g_T, g_P, g_H and time-constant modelling (cool-down) to automatically adjust feedback strength based on previous step efficacy.

In practice, this evolution corresponds to a transition from single-channel pixel feedback (Exp-A/B) to a *multi-channel hybrid estimator* (Exp-D), where both OPL physics and centroid drift contribute to a unified temperature correction.

13.1 OPL Residual: Removing Environmental Bias

The raw IDS measurement of OPL contains both true mechanical drift and refractive-index changes caused by pressure and humidity. Therefore, the OPL is first corrected:

$$\text{OPL}_{\text{residual}}(t) = \text{OPL}(t) - \underbrace{(a_0 + a_1 t)}_{\text{warmup baseline}} - g_P \Delta P(t) - g_H \Delta H(t).$$

This isolates the *true physical drift* of the system. From this, temperature error is estimated via the fitted thermo-optic coefficient g_T ($\mu\text{m}/^\circ\text{C}$):

$$eT_{\text{OPL}} = - \frac{\text{OPL}_{\text{residual}}}{g_T}.$$

13.2 Pixel-Based Temperature Estimate

Rolling regression is used to learn the mapping

$$\begin{bmatrix} \Delta X \\ \Delta Y \end{bmatrix} \approx \boldsymbol{\beta} \cdot \begin{bmatrix} \Delta T \\ \Delta P \\ \Delta H \end{bmatrix}.$$

Only the *temperature component* is used for feedback:

$$eT_{\text{PX}} = - \frac{\boldsymbol{\beta}_T^\top (\vec{d} - \text{P/H terms})}{\|\boldsymbol{\beta}_T\|^2}, \quad \boldsymbol{\beta}_T = (\beta_{X,T}, \beta_{Y,T}).$$

This allows the pixel motion to inform temperature corrections while removing the part that is explainable by pressure and humidity variations.

13.3 OPL $\oplus$ PX Fusion (Hybrid Estimate)

Both channels provide independent estimates of temperature. The final control signal is a weighted fusion:

$$eT_{\text{fused}} = w eT_{\text{OPL}} + (1 - w) eT_{\text{PX}} \quad \text{with} \quad 0.8 \leq w \leq 0.95,$$

where eT_{OPL} represents the physically-derived estimate from OPL residuals, and eT_{PX} is the centroid-based estimate. A higher weight is typically assigned to OPL because it exhibits lower short-term noise and stronger physical validity.

14 Results and Performance Evaluation for experiment H (FINAL)

This section presents a comparative assessment of the closed-loop feedback experiments conducted between November and December 2025, culminating in a long-duration benchmark run (Exp-H). The primary objective is to evaluate the achievable spectrograph image stability using temperature-only feedback under varying environmental conditions and to establish the operational limits of the current control architecture.

The experiment consists of three stages: an initial ~ 1 h calibration phase for acquiring reference and calibration frames, followed by a ~ 1 h warm-up period without feedback during which environmental and optical data are collected to initialise the model, and a final ~ 6 h closed-loop phase with active temperature-based feedback. Across the full experiment, the TEC setpoint spans a total range of $\Delta T \approx 1^\circ\text{C}$, while the environment exhibits substantial variation, including a pressure change of $\Delta P \approx 5$ hPa, a humidity excursion of $\Delta H \approx 2.5\%$, and a corresponding refractive-index variation of order $\Delta n \sim 10^{-6}$. Despite these combined perturbations, image motion remains bounded throughout the closed-loop phase, with ΔX confined to sub-pixel levels and ΔY maintained within a few-pixel envelope, demonstrating stable long-term operation using temperature-only feedback under evolving environmental conditions.

14.1 Comparative Closed-Loop Performance

Tables 2–6 summarise the performance of all experiments (A–H) using pixel stability, optical path length (OPL) residuals, and environmental robustness as evaluation metrics.

Early experiments (Exp-A to Exp-C) demonstrate limited or inconsistent stabilisation, with feedback either ineffective or unable to suppress drift under modest environmental forcing. Intermediate hybrid implementations (Exp-D to Exp-F) show improved short-

Table 2: Pixel stability comparison across all feedback experiments. RMS values are computed separately for warm-up and feedback phases.

Experiment	RMS ΔY		RMS ΔX		Duration	Class	Assessment
	Warm-up (px)	Feedback (px)	Feedback (px)				
Exp-A	2.92	8.53	0.83	1.0 h	Short		Unstable
Exp-B	4.28	4.88	2.06	1.8 h	Short		Weak control
Exp-C	1.88	3.06	0.95	1.0 h	Short		Partial control
Exp-D	1.19	1.78	0.29	4.1 h	Medium		Moderate stability
Exp-Ea	0.24	0.48	0.12	1.6 h	Short		Best short-run
Exp-Eb	0.66	2.13	0.29	2.3 h	Medium		Environment-limited
Exp-F	0.88	1.59	0.25	4.9 h	Medium		Drift-prone
Exp-G	0.65	1.22	0.10	3.3 h	Medium		Fragmented run
Exp-H	0.79	0.83	0.16	8.0 h	Long		Best long-run stability

Table 3: Optical path length (OPL) residual stability during warm-up and feedback phases.

Experiment	RMS OPL _{warm} [μm]	RMS OPL _{fb} [μm]	Interpretation
Exp-A	0.80	0.96	No correction
Exp-B	0.52	3.89	OPL instability
Exp-C	1.70	2.29	Partial correction
Exp-D	0.71	0.54	Good suppression
Exp-Ea	0.07	0.53	Excellent short run
Exp-Eb	0.37	0.84	Mixed
Exp-F	0.36	1.04	Environment-driven
Exp-G	0.95	1.08	Fragmented behaviour
Exp-H	0.77	6.47	RI-limited (expected)

term stability, particularly in reduced RMS pixel drift during feedback phases; however, residual drift remains correlated with pressure and humidity variations.

Exp-Ea achieves the lowest absolute RMS values but over a relatively short duration and under comparatively mild environmental conditions. Exp-G demonstrates good pixel stability.

In contrast, Exp-H provides the most stringent test of the control system, combining extended duration, frequent feedback actions, and the largest environmental excursions observed across all experiments.

14.2 Long-Run Stability Benchmark: Exp-H (Hybrid OPL $\oplus$ PX)

Experiment H represents the final and most comprehensive evaluation of the hybrid OPL $\oplus$ pixel feedback architecture. The experiment was conducted over approximately eight hours, during which the system experienced significant atmospheric pressure ($\Delta P \approx 5.1$ hPa) and humidity ($\Delta H \approx 2.8\%$) variations, while temperature was the sole controlled actuator.

Table 4: Environmental ranges experienced during each experiment.

Experiment	ΔT [°C]	ΔP [hPa]	ΔH [%]	Robustness
Exp-A	0.01	0.04	0.13	Trivial
Exp-B	1.03	0.15	1.90	Unstable
Exp-C	1.02	0.58	0.45	Moderate
Exp-D	0.43	0.78	0.66	Robust
Exp-Ea	0.54	0.49	0.29	Controlled
Exp-Eb	0.69	0.84	0.46	Stress-tested
Exp-F	0.17	0.94	0.95	AO-limited
Exp-G	0.52	0.17	0.62	Moderate
Exp-H	1.10	5.14	2.79	Best stress test

Table 5: Summary of long-run closed-loop performance for Experiment H.

Metric	Value
Duration	~ 8 hours
Feedback steps	19
Total applied $\sum \Delta T $	1880 mK
RMS ΔY_{warm}	0.786 px
RMS ΔY_{fb}	0.832 px
RMS ΔX_{fb}	0.161 px
RMS OPL _{fb}	6.47 μm
$\text{corr}(\Delta Y, T_{\text{meas}})$	+0.71
$\text{corr}(\Delta Y, \Delta P)$	+0.58
$\text{corr}(\Delta Y, \Delta H)$	-0.53
$\text{corr}(\Delta Y, \text{RI})$	+0.11

Despite these conditions, the RMS pixel drift during the feedback phase remains comparable to warm-up values, with RMS $\Delta Y_{\text{fb}} = 0.83$ px and RMS $\Delta X_{\text{fb}} = 0.16$ px. Importantly, no evidence of runaway drift or oscillatory behaviour is observed, even as the cumulative applied temperature correction exceeds 1.8 K.

Temporal stability metrics further highlight the robustness of this run. Exp-H exhibits the lowest drift rates across all experiments, with RMS $\Delta Y = 0.022$ px min⁻¹ and RMS $\Delta X = 0.010$ px min⁻¹, representing an order-of-magnitude improvement over early hybrid implementations.

The elevated OPL residuals observed during this run are consistent with uncorrected refractive index variations driven by atmospheric pressure and humidity. Correlation analysis confirms that while pixel drift remains strongly correlated with measured temperature ($r \approx 0.71$), residual sensitivity to pressure and humidity persists. This behaviour is expected given the absence of active correction for these parameters within the present control loop.

14.3 Control Limit and Implications

Experiment H establishes the practical performance boundary of temperature-only closed-loop stabilisation for the current spectrograph configuration. Over an uninterrupted ~ 8 hour run, the controller maintained bounded image motion despite substantial environmental evolution, providing a quantitative measure of the limits of thermal actuation.

During Exp-H, the feedback loop constrained the primary dispersion-axis motion to an RMS $\Delta Y_{\text{fb}} = 0.83$ px, comparable to the warm-up baseline (0.79 px), while the orthogonal axis remained stable at RMS $\Delta X_{\text{fb}} = 0.16$ px. This indicates that first-order thermally driven image drift is effectively suppressed over long timescales, even under sustained external forcing. The temporal stability achieved corresponds to RMS drift rates of $0.022 \text{ px min}^{-1}$ (1.32 px hr^{-1}) in ΔY and $0.010 \text{ px min}^{-1}$ (0.63 px hr^{-1}) in ΔX , representing the best long-run performance across all experiments.

Crucially, this stability was achieved under significantly stronger environmental variation than in earlier runs. Over the full duration of Exp-H, the internal air temperature varied by $\Delta T_{\text{env}} \simeq 1.10^\circ\text{C}$, pressure by $\Delta P_{\text{env}} \simeq 5.14 \text{ hPa}$, and relative humidity by $\Delta H_{\text{env}} \simeq 2.79\%$. Despite these changes, image motion remained bounded and non-divergent, demonstrating that the thermal loop can tolerate pressure excursions of order $\sim 5 \text{ hPa}$ and humidity variations of order $\sim 3\%$ without loss of image stability.

However, the experiment also exposes the intrinsic limitation of temperature-only control. While pixel drift remained constrained, the optical path length residual increased to RMS $\text{OPL}_{\text{res,fb}} = 6.47 \text{ }\mu\text{m}$, substantially larger than in shorter, more benign runs. Correlation analysis shows that residual ΔY during feedback retains measurable sensitivity to environmental parameters, with $\text{corr}(\Delta Y, P_{\text{env}}) \approx +0.58$ and $\text{corr}(\Delta Y, H_{\text{env}}) \approx -0.53$, indicating refractive-index-driven image motion that cannot be compensated through thermal actuation alone.

Taken together, Exp-H defines the long-run operating envelope of the current control architecture. Temperature feedback can reliably stabilise image motion to the sub-pixel level over multi-hour timescales, even under $\sim 1^\circ\text{C}$ temperature swings and multi-hPa pressure evolution. Beyond this regime, residual image motion is dominated by refractive index variations driven by pressure and humidity, rather than by thermal drift of the opto-mechanical structure.

In this sense, Exp-H represents the maximum achievable precision using temperature-only feedback and provides a quantitative baseline for future upgrades. The observed residual correlations motivate the introduction of active optics as a complementary control layer, capable of directly correcting refractive-index-induced image motion that lies outside the authority of thermal control. Subsequent active-optics-assisted experiments can therefore be evaluated directly against the performance envelope established by Exp-H.

Table 6: Temporal pixel stability and environmental disturbance across all experiments.

Experiment	RMS ΔY (px/min)	RMS ΔY (px/hr)	RMS ΔX (px/min)	RMS ΔX (px/hr)	ΔT
Exp-A	0.236	14.16	0.109	6.52	
Exp-B	0.413	24.77	0.171	10.26	
Exp-C	0.286	17.18	0.118	7.07	
Exp-D	0.075	4.51	0.088	5.30	
Exp-Ea	0.075	4.48	0.087	5.21	
Exp-Eb	0.094	5.65	0.076	4.58	
Exp-F	0.067	4.05	0.075	4.48	
Exp-G	0.055	3.28	0.015	0.92	
Exp-H	0.022	1.32	0.010	0.63	

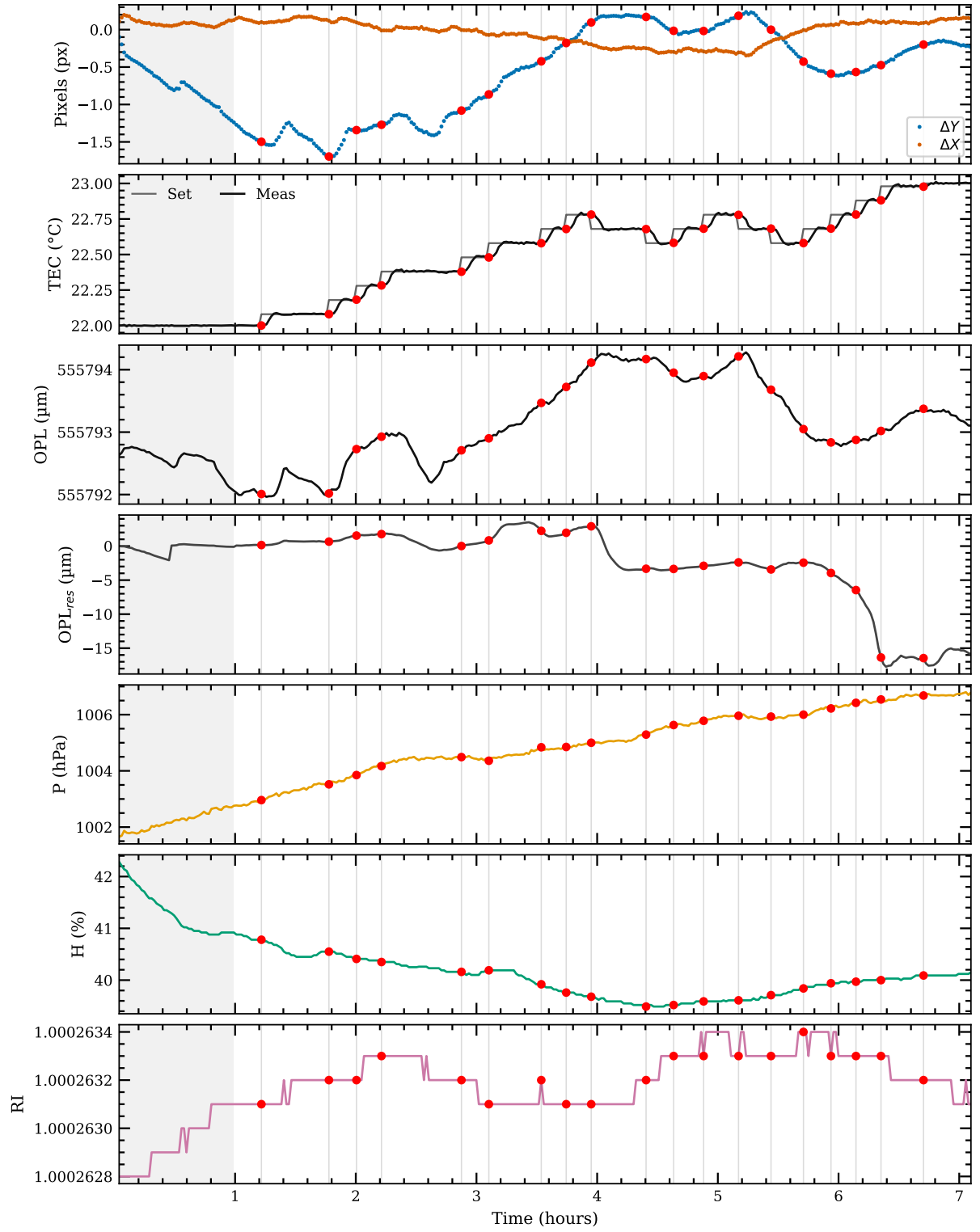


Fig. 7: Long-duration closed-loop performance of Experiment H (Hybrid OPL $\oplus$ PX) over an ~ 8 h run. Top to bottom: (a) pixel drift in the cross-dispersion (ΔY) and dispersion (ΔX) directions, (b) TEC setpoint and measured temperature, (c) optical path length (OPL), (d) OPL residual after baseline removal, (e) atmospheric pressure, (f) relative humidity, and (g) refractive index derived from ECU measurements. Red markers indicate applied feedback updates.